

A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-35663739})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K , the normalized generator σ_x , the pair (a', J) representing the torsion class e_j , the auxiliary divisor I' , the compactly represented element t , a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$(1 - \sigma_x)^2 I' + \text{div}(t) + i_x(J) = 0, \quad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$ -basis of the ideal. For example, the ideal $J = \begin{pmatrix} 3255 & 1924 \\ 0 & 1 \end{pmatrix}$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z} 3255 + \mathbf{Z} (1924 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\text{Cl}(K)$ (as HNF ideals). A character x on $\text{Cl}(K)/5$ is recorded by its value vector $(x(\bar{\kappa}_1), x(\bar{\kappa}_2), x(\bar{\kappa}_3))$; thus $(1, 0, 0)$ is the character dual to $\bar{\kappa}_1$. The column number j records which basis element e_j of $\text{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x . Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x .

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 - s + 8915935 = 0$, of discriminant -35663739 . Class group invariants $(30 \ 10 \ 5)$, class number 1500; the three stored generator ideals (HNF) are $\begin{pmatrix} 137 & 94 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 1605 & 490 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 655 & 269 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a , column e_1

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 332y^3 + (2s - 1)y^2 + 173502y + (1224s - 612)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 664y^8 + 457228y^6 + 150869067y^4 + 73755360540y^2 + 13357639460016$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{440032}{1159963732423125} + \left(\frac{21893}{40598730634809375}\right)s, \quad J = \begin{pmatrix} 3255 & 1924 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3255$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $10565306426304 = 2^6 \cdot 3^2 \cdot 7^2 \cdot 17 \cdot 313 \cdot 70351$:

$$\begin{pmatrix} 31444364364 & 20202387876 & 12851504618 & 8259209223 & 15120591628 & 6779836833 & 9141892896 & 14037702504 & 18289279276 & 29881837590 \\ 0 & 12 & 2 & 1 & 4 & 8 & 5 & 11 & 5 & 3 \\ 0 & 0 & 14 & 3 & 2 & 11 & 1 & 10 & 1 & 13 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 197 factors with signed exponents of absolute value up to 1576961670; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 0 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 2: character a , column e_2

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 332y^3 + (2s - 1)y^2 + 173502y + (1224s - 612)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 664y^8 + 457228y^6 + 150869067y^4 + 73755360540y^2 + 13357639460016$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{1879447}{104074241381523} + \left(-\frac{949}{104074241381523}\right)s, \quad J = \begin{pmatrix} 987 & 238 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 987$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $20802457174464 = 2^6 \cdot 3^5 \cdot 7^2 \cdot 197 \cdot 138569$:

$$\begin{pmatrix} 2293039812 & 1523791752 & 26025892 & 1247666655 & 1397914383 & 67341697 & 2024433285 & 1971708712 & 1755460136 & 2265108207 \\ 0 & 42 & 18 & 27 & 27 & 24 & 21 & 20 & 23 & 32 \\ 0 & 0 & 4 & 3 & 3 & 3 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 3 & 0 & 0 & 2 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 191 factors with signed exponents of absolute value up to 57315; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \ 3 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 3: character a , column e_3

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 332y^3 + (2s - 1)y^2 + 173502y + (1224s - 612)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 664y^8 + 457228y^6 + 150869067y^4 + 73755360540y^2 + 13357639460016$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1695709}{24112181031875} + \left(-\frac{2337}{120560905159375}\right) s, \quad J = \begin{pmatrix} 655 & 269 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 655$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2917152748908 = 2^2 \cdot 3^4 \cdot 17 \cdot 47 \cdot 1171 \cdot 9623$:

$$\begin{pmatrix} 162064041606 & 137019750354 & 8473130983 & 52420887168 & 67359498522 & 79448464693 & 122571838641 & 18593897230 & 97785198228 & 8468098328 \\ 0 & 6 & 3 & 3 & 3 & 3 & 2 & 3 & 4 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 1 & 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 188 factors with signed exponents of absolute value up to 1060; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 4: character b , column e_1

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 1553y^3 + (8s - 4)y^2 + 2340789y + (-9576s + 4788)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 3106y^8 + 7093387y^6 + 7841110458y^4 + 4113229283865y^2 + 817589219405616$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \ -1 \ 2 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{440032}{1159963732423125} + \left(\frac{21893}{40598730634809375}\right) s, \quad J = \begin{pmatrix} 3255 & 1924 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3255$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $25036491035817 = 3^3 \cdot 107 \cdot 409 \cdot 21188617$:

$$\begin{pmatrix} 2781832337313 & 2077812088608 & 1360140153417 & 2366298449845 & 331118006177 & 1129823163590 & 59528349723 & 562953834221 & 1456307020278 & 2005627194803 \\ 0 & 3 & 0 & 1 & 2 & 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 3 & 2 & 1 & 2 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 250 factors with signed exponents of absolute value up to 119708697199613794561605; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \ 0 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 5: character b , column e_2

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 1553y^3 + (8s - 4)y^2 + 2340789y + (-9576s + 4788)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 3106y^8 + 7093387y^6 + 7841110458y^4 + 4113229283865y^2 + 817589219405616$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \ -1 \ 2 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{1879447}{104074241381523} + \left(-\frac{949}{104074241381523}\right)s, \quad J = \begin{pmatrix} 987 & 238 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 987$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2665698605244 = 2^2 \cdot 3^6 \cdot 10789 \cdot 84731$:

$$\begin{pmatrix} 16454929662 & 9971860647 & 8587602123 & 6984204257 & 9048264159 & 13112733836 & 3530574461 & 7780144140 & 7643174006 & 9960270623 \\ 0 & 3 & 0 & 2 & 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 3 & 1 & 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 & 1 & 0 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 3 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 245 factors with signed exponents of absolute value up to 51923777674526990740550; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_2) = (2 \ 0 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 6: character b , column e_3

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 1553y^3 + (8s - 4)y^2 + 2340789y + (-9576s + 4788)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 3106y^8 + 7093387y^6 + 7841110458y^4 + 4113229283865y^2 + 817589219405616$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \ -1 \ 2 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1695709}{24112181031875} + \left(-\frac{2337}{120560905159375}\right)s, \quad J = \begin{pmatrix} 655 & 269 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 655$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $504473186448 = 2^4 \cdot 3^2 \cdot 7^2 \cdot 11^2 \cdot 67 \cdot 8819$:

$$\begin{pmatrix} 545966652 & 240672993 & 13897140 & 239940997 & 373501409 & 370017632 & 540437499 & 532675738 & 455483489 & 102061283 \\ 0 & 33 & 11 & 26 & 11 & 11 & 18 & 18 & 17 & 32 \\ 0 & 0 & 7 & 1 & 1 & 3 & 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 4 & 0 & 0 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 244 factors with signed exponents of absolute value up to 43458523625733493475426; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \ 0 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 7: character c , column e_1

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + (-s - 79)y^3 + (34s - 38164)y^2 + (-475s + 951764)y + (2442s - 8605014)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 159y^8 - 76294y^7 + 10825308y^6 - 617425776y^5 + 20080859 \\ & 479y^4 - 402757877448y^3 + 5054022702159y^2 - 37057551106554y + \\ & 127194218301348 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 0 \ 0 \ 0 \ -1 \ 0 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{440032}{1159963732423125} + \left(\frac{21893}{40598730634809375} \right) s, \quad J = \begin{pmatrix} 3255 & 1924 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3255$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $7736623235829 = 3^6 \cdot 10612651901$:

$$\begin{pmatrix} 95513867109 & 35395789323 & 13201038366 & 46711363500 & 12949911567 & 16406697115 & 32289976453 & 80650029660 & 59050582979 & 81915923263 \\ 0 & 3 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 3 & 0 & 0 & 2 & 1 & 0 & 1 & 2 \\ 0 & 0 & 0 & 3 & 0 & 1 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 3 & 2 & 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 260 factors with signed exponents of absolute value up to 12342809; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (2 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 8: character c , column e_2

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + (-s - 79)y^3 + (34s - 38164)y^2 + (-475s + 951764)y + (2442s - 8605014)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 159y^8 - 76294y^7 + 10825308y^6 - 617425776y^5 + 20080859 \\ & 479y^4 - 402757877448y^3 + 5054022702159y^2 - 37057551106554y + \\ & 127194218301348 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 0 \ 0 \ 0 \ -1 \ 0 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{1879447}{104074241381523} + \left(-\frac{949}{104074241381523}\right)s, \quad J = \begin{pmatrix} 987 & 238 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 987$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $73658776572 = 2^2 \cdot 3^4 \cdot 43 \cdot 5287021$:

$$\begin{pmatrix} 1364051418 & 18911010 & 167500668 & 358584765 & 313568239 & 618976896 & 1358953953 & 219145074 & 637273525 & 1031816654 \\ 0 & 6 & 0 & 3 & 4 & 5 & 5 & 3 & 1 & 4 \\ 0 & 0 & 3 & 0 & 1 & 2 & 2 & 2 & 2 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 255 factors with signed exponents of absolute value up to 1882221859; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \ 1 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 9: character c , column e_3

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + (-s - 79)y^3 + (34s - 38164)y^2 + (-475s + 951764)y + (2442s - 8605014)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 159y^8 - 76294y^7 + 10825308y^6 - 617425776y^5 + 20080859 \\ & 479y^4 - 402757877448y^3 + 5054022702159y^2 - 37057551106554y + \\ & 127194218301348 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 0 \ 0 \ 0 \ -1 \ 0 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1695709}{24112181031875} + \left(-\frac{2337}{120560905159375}\right)s, \quad J = \begin{pmatrix} 655 & 269 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 655$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $3456311578704 = 2^4 \cdot 3^3 \cdot 1571 \cdot 5092757$:

$$\begin{pmatrix} 96008654964 & 38110457248 & 18668230464 & 21763382877 & 27962968834 & 23439492850 & 12875245230 & 8233796532 & 53052385570 & 3048570973 \\ 0 & 4 & 0 & 3 & 1 & 1 & 2 & 3 & 2 & 1 \\ 0 & 0 & 3 & 0 & 0 & 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 & 1 & 2 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 255 factors with signed exponents of absolute value up to 2974198877; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \ 4 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + 718y^3 + (24s + 75601)y^2 + (426s - 1259150)y + (-252s - 23763152)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 + 1440y^8 + 148354y^7 - 2304802y^6 + 66089460y^5 + \\ & 9140123765y^4 - 42195040202y^3 - 498522080904y^2 + 57918652976408 \\ & *y + 565259578825648 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{440032}{1159963732423125} + \left(\frac{21893}{40598730634809375} \right) s, \quad J = \begin{pmatrix} 3255 & 1924 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3255$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $9025762339584 = 2^8 \cdot 3^2 \cdot 19^2 \cdot 10851611$:

$$\begin{pmatrix} 2474167308 & 1164074292 & 184340060 & 359135148 & 2315003740 & 2301188020 & 1215514084 & 2276873455 & 1165896335 & 647190079 \\ 0 & 76 & 64 & 0 & 53 & 60 & 7 & 74 & 43 & 39 \\ 0 & 0 & 4 & 0 & 1 & 3 & 0 & 3 & 0 & 1 \\ 0 & 0 & 0 & 12 & 0 & 1 & 0 & 1 & 0 & 4 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 177 factors with signed exponents of absolute value up to 187471390; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \ 4 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 11: character $a + b$, column e_2

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + 718y^3 + (24s + 75601)y^2 + (426s - 1259150)y + (-252s - 23763152)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 + 1440y^8 + 148354y^7 - 2304802y^6 + 66089460y^5 + \\ & 9140123765y^4 - 42195040202y^3 - 498522080904y^2 + 57918652976408 \\ & *y + 565259578825648 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{1879447}{104074241381523} + \left(-\frac{949}{104074241381523}\right)s, \quad J = \begin{pmatrix} 987 & 238 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 987$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2136229565904 = 2^4 \cdot 3^3 \cdot 7 \cdot 41^2 \cdot 420241$:

$$\begin{pmatrix} 723655002 & 574057044 & 235636758 & 649008388 & 80605386 & 509071573 & 109248393 & 17128073 & 623091001 & 685243890 \\ 0 & 6 & 0 & 1 & 0 & 0 & 5 & 1 & 2 & 1 \\ 0 & 0 & 246 & 209 & 84 & 231 & 1 & 68 & 73 & 234 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 174 factors with signed exponents of absolute value up to 267645946; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_2) = (4 \ 1 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 12: character $a + b$, column e_3

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + 718y^3 + (24s + 75601)y^2 + (426s - 1259150)y + (-252s - 23763152)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 4y^9 + 1440y^8 + 148354y^7 - 2304802y^6 + 66089460y^5 + 9140123765y^4 - 42195040202y^3 - 498522080904y^2 + 57918652976408y + 565259578825648$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1695709}{24112181031875} + \left(-\frac{2337}{120560905159375}\right)s, \quad J = \begin{pmatrix} 655 & 269 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 655$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1790353844307 = 3^3 \cdot 7 \cdot 449 \cdot 21097487$:

$$\begin{pmatrix} 198928204923 & 182752009827 & 192410132604 & 101051000233 & 31094290314 & 72156000094 & 21871988757 & 155446740482 & 56780140274 & 115519788615 \\ 0 & 3 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 & 1 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 171 factors with signed exponents of absolute value up to 495160; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \ 2 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 323y^3 + (-8s + 4)y^2 + 682920y + (7632s - 3816)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 646y^8 + 1470169y^6 + 1011786144y^4 - 622362897792y^2 + 519330231739584$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ -1 \ 1 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{440032}{1159963732423125} + \left(\frac{21893}{40598730634809375}\right) s, \quad J = \begin{pmatrix} 3255 & 1924 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3255$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1427698351524 = 2^2 \cdot 3 \cdot 17 \cdot 6998521331$:

$$\begin{pmatrix} 713849175762 & 309742215377 & 356881487816 & 310077959300 & 453006801053 & 365069289013 & 689983809939 & 84365081848 & 429588857344 & 336843479018 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = \begin{pmatrix} 3 & 3 & 2 \end{pmatrix}$ in the coordinates of $\text{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $\begin{pmatrix} 1 & 0 & 1 \end{pmatrix}$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 323y^3 + (-8s + 4)y^2 + 682920y + (7632s - 3816)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 646y^8 + 1470169y^6 + 1011786144y^4 - 622362897792y^2 + 519330231739584$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ -1 \ 1 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{1879447}{104074241381523} + \left(-\frac{949}{104074241381523}\right) s, \quad J = \begin{pmatrix} 987 & 238 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 987$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $204632784 = 2^4 \cdot 3^3 \cdot 151 \cdot 3137$:

$$\begin{pmatrix} 2842122 & 110097 & 663606 & 1640288 & 2756777 & 2640022 & 26614 & 694045 & 1631768 & 2587573 \\ 0 & 3 & 0 & 0 & 2 & 0 & 2 & 1 & 2 & 2 \\ 0 & 0 & 6 & 3 & 1 & 4 & 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 332 factors with signed exponents of absolute value up to 78011678501730645021338488331515; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_2) = (1 \ 2 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 323y^3 + (-8s + 4)y^2 + 682920y + (7632s - 3816)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 646y^8 + 1470169y^6 + 1011786144y^4 - 622362897792y^2 + 519330231739584$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ -1 \ 1 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1695709}{24112181031875} + \left(-\frac{2337}{120560905159375}\right)s, \quad J = \begin{pmatrix} 655 & 269 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 655$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $587918327652 = 2^2 \cdot 3^3 \cdot 33223 \cdot 163853$:

$$\begin{pmatrix} 32662129314 & 4811323227 & 8541670383 & 28694935295 & 8771928401 & 12623816159 & 14994032186 & 18190866962 & 19896138292 & 13620890396 \\ 0 & 3 & 0 & 1 & 0 & 2 & 1 & 0 & 2 & 0 \\ 0 & 0 & 3 & 2 & 0 & 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 332 factors with signed exponents of absolute value up to 42915941552028417783044070716172; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \ 4 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 173y^3 + (6s - 3)y^2 - 333198y + (-2214s + 1107)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 346y^8 - 636467y^6 + 205687143y^4 - 125857647234y^2 + 43704093293811$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ 0 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{440032}{1159963732423125} + \left(\frac{21893}{40598730634809375} \right) s, \quad J = \begin{pmatrix} 3255 & 1924 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3255$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $89913145003200 = 2^6 \cdot 3^5 \cdot 5^2 \cdot 17^2 \cdot 181 \cdot 4421$:

$$\begin{pmatrix} 2448615060 & 2054213640 & 2058072050 & 2432657719 & 1693970510 & 944214447 & 815236219 & 818352107 & 2295049204 & 831016198 \\ 0 & 60 & 4 & 35 & 48 & 29 & 42 & 37 & 39 & 2 \\ 0 & 0 & 34 & 5 & 18 & 19 & 22 & 31 & 13 & 4 \\ 0 & 0 & 0 & 3 & 0 & 1 & 2 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 182 factors with signed exponents of absolute value up to 11959071; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 4 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 173y^3 + (6s - 3)y^2 - 333198y + (-2214s + 1107)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 346y^8 - 636467y^6 + 205687143y^4 - 125857647234y^2 + 43704093293811$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ 0 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{1879447}{104074241381523} + \left(-\frac{949}{104074241381523}\right)s, \quad J = \begin{pmatrix} 987 & 238 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 987$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $89913145003200 = 2^6 \cdot 3^5 \cdot 5^2 \cdot 17^2 \cdot 181 \cdot 4421$:

$$\begin{pmatrix} 2448615060 & 2054213640 & 2058072050 & 2432657719 & 1693970510 & 944214447 & 815236219 & 818352107 & 2295049204 & 831016198 \\ 0 & 60 & 4 & 35 & 48 & 29 & 42 & 37 & 39 & 2 \\ 0 & 0 & 34 & 5 & 18 & 19 & 22 & 31 & 13 & 4 \\ 0 & 0 & 0 & 3 & 0 & 1 & 2 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 189 factors with signed exponents of absolute value up to 11876792; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \ 4 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 173y^3 + (6s - 3)y^2 - 333198y + (-2214s + 1107)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 346y^8 - 636467y^6 + 205687143y^4 - 125857647234y^2 + 43704093293811$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ 0 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1695709}{24112181031875} + \left(-\frac{2337}{120560905159375}\right)s, \quad J = \begin{pmatrix} 655 & 269 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 655$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $762042390000 = 2^4 \cdot 3 \cdot 5^4 \cdot 911 \cdot 27883$:

$$\begin{pmatrix} 3810211950 & 2324041530 & 3611478250 & 1040602110 & 3697042603 & 2607704572 & 671968116 & 2260715024 & 707808867 & 3473898188 \\ 0 & 10 & 0 & 4 & 4 & 5 & 0 & 6 & 2 & 6 \\ 0 & 0 & 5 & 2 & 2 & 0 & 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & 2 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 195 factors with signed exponents of absolute value up to 259033; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \ 3 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.